Green Synthesis Approach of Lignin Nanoparticles, Micro Alumino-silicate and Micro-Silica from Rice Husk via Soda Pulping Method

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Highlights

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Conversion of Rice husk into Lignin nanoparticles is a value-add process.

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Lignin nanoparticles (LNPs) are biocompatible and can be made using different methods.

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Rice husk was used to produce LNPs and silica nanoparticles through a soda pulping process.

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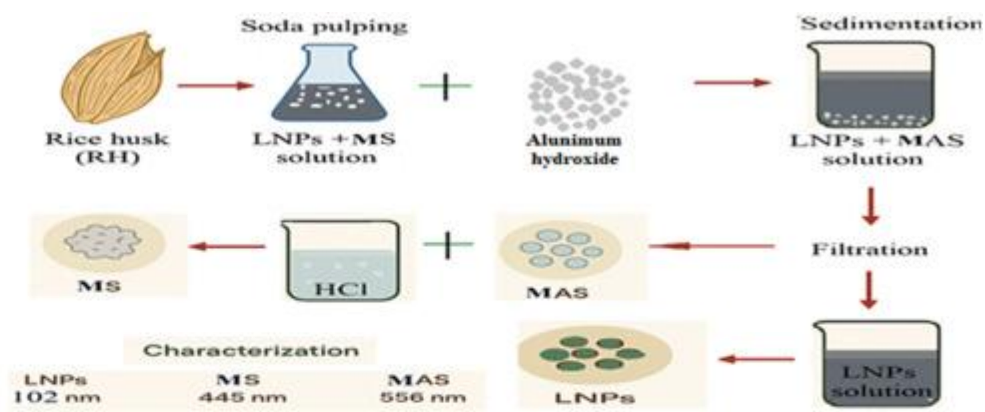
The isolation of LNPs involved a reaction with aluminium to create Micro-Alumino-silicate, and products were analyzed for particle sizes.


Abstract

Biomass, defined as organic material derived from plant or animal sources, represents a renewable feedstock for value-added materials. In this study, rice husk (RH), an abundant agricultural residue, as a renewable source for producing lignin nanoparticles (LNPs), Micro Alumino-silicate (MAS), and Micro-silica (MS). The process begins with soda pulping of rice husk, which simultaneously extracts lignin nanoparticles and generates a silica-rich phase. This process yielded a colloidal system containing dispersed LNPs and MS. A novel Aluminum-assisted precipitation strategy was subsequently developed to isolate LNPs efficiently. In this approach, MS was chemically converted into MAS via reaction with Aluminum hydroxide ($Al(OH)_3$), inducing rapid precipitation and enabling separation of LNPs from the suspension. Notably, this method enables effective lignin recovery without measurable lignin loss, thereby preserving biomass carbon efficiency. Conducted under mild conditions, contributing to low energy consumption, while the controlled use of Aluminum-based reagents minimizes chemical demand. Furthermore, the reliance on conventional unit operations supports the scalability of the method for industrial applications. The obtained products (LNPs, MS, and MAS) were characterized using Fourier transform infrared spectroscopy (FTIR), particle size analysis (PSA), and scanning electron microscopy (SEM). Particle size distribution analysis revealed.


Graphical abstract

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Introduction

Biopolymer nanoparticles (BPn) can be separated into two categories: natural (polysaccharide- and protein-based) and synthetic [1]. Several techniques have been reported for the preparation of nanoparticles , each of which produces nanoparticles with a unique shape and surface charge [2]. These techniques include emulsification, ionic gelation, electrospray drying, and nanoprecipitation [3]. The characteristics and utilization of BPNs are dependent upon their morphology. [4]

Lignin is a biopolymer found mainly in the cell walls of lignocellulosic biomass and includes matter such as wood, grass, and crop stalks. It is the second most common polymer in nature, immediately after cellulose. [5] Agricultural and industrial biomass, both of which are sources of lignin, pollute the environment when they are discarded. Compared to that of other biomaterials, the price of lignin is relatively low because most of its cost is associated with obtaining it at an extraction facility.[6], [7]

The lignin structure is sensitive to growth conditions, biomass type, and extraction methods [6], [8]. Because of its many beneficial properties, including biodegradability, biocompatibility, antioxidant activity, lack of cytotoxicity, and ability to increase soil nutrients through microbial degradation, lignin is an ideal choice for use as a precursor in the creation of eco-friendly nanoscale materials.[9], [10]

Sinapyl alcohol, coniferyl alcohol, and coumaryl alcohol are the three phenylpropane monomers that constitute the bulk of the lignin structure, regardless of the plant or extraction method. [11], [12] The linkage of primary and secondary metabolism in plants leads to the formation of different materials [13], [14], [15]. In fact, primary carbon metabolism results in the formation of proteins, lipids, carbohydrates, and chlorophyll. [16], [17] Similarly, secondary carbohydrate metabolism results in the production of terpenoids, alkaloids, cyanogenic glycosides, and phenolic compounds. [18]

The use of renewable resource feedstocks in the chemical industry and in academic chemistry laboratories is encouraged by the green chemistry field [19], [20]. Only 40% of lignin is used for biorefining because using it exclusively for energy generation is not profitable [21]. According to estimates, lignin production in 2025 will reach approximately $913 million per year. [22] To improve the bioeconomy, novel and value-added applications of lignin must be investigated to increase the profitability of relevant biorefinery processes [23]. In particular, the pulp and paper industry generates more than 30 million tons a year of waste, much of which is burned to create electricity. One of the world's foremost producers of chemicals derived from lignin is China, followed by India, Japan, and Indonesia.[22]

The type of lignin used varies depending on the extraction method used for lignocellulosic biomass. The sulphur and silica contents affect the purity of the resulting lignin [24]. For example, the sulfur and silica contents are low in Kraft pulp, resulting in high-purity lignin. In contrast, the low purity of lignosulfonate is caused by the high content of sulfur and silica. Various lignin types have been processed at both the industrial and pilot scales to achieve a high production rate. Additionally, a life cycle assessment (LCA) of the development of novel production methods for lignin nanomaterials supports a circular economy at the technical and economic levels [23], [25]. Research into LCA has shown that incorporating lignin into product production has the potential to improve environmental performance across a variety of environmental impact categories, including carbon footprints, climate change, and greenhouse gas emissions. [22]

At the technical and economic levels, lignin can be biorefined and turned into products with added value.[23], [26] Lignin nanoparticles (LNPs) have been the subject of extensive study in the field of functional composites in recent years as a result of their wide range of beneficial properties, including antimicrobial, wastewater treatment [27], and UV-absorbing properties. Many ground-breaking efforts are currently focused on expanding the utility of LNP-based composites.[28] Intensive efforts have been devoted in recent decades to investigating and utilizing lignin, resulting in a vast expansion of lignin applications [29]. Since lignin is obtained from industrial and agricultural waste and its use is supported by green chemistry [30], [31], transforming lignin into LNPs and employing it in technology and engineering may represent a step toward global sustainability [32]. The most efficient way to recycle the massive amounts of biowastes produced by agriculture and industry is through the preparation of bio-based materials and their numerous high-tech applications [33]. Additionally, Lignin serves as a promising alternative raw material for phenol due to its structural similarity with benzene propane units. [34]

There has been a trend in research into synthesizing LNPs, and several different approaches, such as $CO_2$ saturation, solvent exchange of pure lignin, acetylation of lignin and dialysis, flash-precipitation methods [35], ultrasonication [36], ionic gelation [37], and water-in-oil microemulsion methods, have been tested [38]. Both the lignin precursor and nanoparticle synthesis methods influence the physical properties of the resulting particles, as the wide range of surface properties of LNPs makes them suitable for several eco-friendly uses [35]. The acid precipitation method is one of the simplest and most affordable choices. A hydrothermal curing process is suggested to fabricate lignin functionalized nanomaterials employed to enhance their properties [39].

Goliszek-Chabros et al. [40] utilized lignin nanoparticles, which were previously extracted from spruce and Eucalyptus as sustainable sunscreen additives. The nanoparticles improved UV protection, with spruce showing better results. They also enhanced emulsion stability and rheological properties, indicating their potential as eco-friendly additives.

Researchers report on the synthesis and characterization of a liquid lignin-based photopolymer and its application in additive manufacturing (AM) [41].

F. Lu et al. [42] synthesized alkali lignin (AL) with a low molecular weight, high phenolic hydroxyl content, and good homogeneity by dissolving AL in 80% methanol. The novel LNPs are the ideal size, have high encapsulation efficiency, release ibuprofen in a predictable pattern when exposed to intestinal fluid in vitro, and are gastritis and acid. The methanol-fractionated alkali lignin (MAL) was quaternized, and the LNPs were assembled in an ethanol/water solution. To test how well their hydrophobic oral drug delivery system works, they used ibuprofen (IBU) as a drug.

Researchers introduced a sustainable and efficient approach for the synthesis of hydrophobic LNPs, thereby facilitating the broadening of applications for lignin-based functional composite film materials. [43]

Thalakkale Veettil et al. [44] developed a method to create lignin photonic glasses using ethanol as a green solvent. Lignin nanoparticles can range from 100 nm to 300 nm, producing colors from violet to brown-red with high yields. This method supports scaling up production while following green chemistry principles.

Pourmoazzen et al. [45] synthesized nanoparticles of Kraft lignin modified with cholesteryl chloroformate (Chol. Cl) and ranged in size from 200 to 500 nm. The modified LNPs exhibited high hydrophobicity and could be exploited for the delivery of folic acid. A novel biomaterial is produced via the reaction of lignin with cholesterol, and it exhibits exceptional interfacial and cargo-carrying properties. Its potential use in blends with (nonpolar) synthetic polymers is indicated by the fact that, upon suspension in water, it disperses into nanoparticles with a melting point not present in the original lignin. Preliminary evidence of pH-controlled drug delivery was also found for the hydrophobic modified lignin.

Lignin from coconut husk was used for the first time as a natural UV protector. Two samples, M1 and M2, showed high protection against UV rays, especially when mixed with commercial cream. M2, lighter in color than M1, increased the sun protection factor (SPF) of sunscreens from 20.0 to 36.1, highlighting its potential in cosmetics. [46]

H. Zhang et al. [47] prepared lignin nanoparticles (LNPs) by self-assembly using a choline chloride/lactic acid-based deep eutectic solvent (DES). High yields of 85.6% and purities of 97.8% were achieved when lignin fractions were isolated from corncob residue via effective pretreatment. In addition, the LNPs had a hydrodynamic diameter of 48.66 nm, a polydispersity index less than 0.2, and a zeta potential greater than 42.2 mV.

A new method was developed to create (LNPs) that are very hydrophobic using a ternary deep eutectic solution (DES) and water. These LNPs have fewer hydrophilic groups on their surface, resulting in a zeta potential of -4.9 mV and a water contact angle of up to 140.0°. LNPs were then used as fillers in polyvinyl alcohol (PVA) films, which showed improved hydrophobicity, mechanical strength, UV protection, and biodegradability. [43]

Gai et al. [48] (LNPs) were made using a high-speed homogenizer and combined with cationic starch to improve the paper's barrier performance for packaging. The LNPs created a stable water dispersion and worked well with cationic starch, giving the paper good hydrophobic and resistance properties. This new approach allows for high-barrier, eco-friendly packaging materials.

Bertolo et al. [49] utilized sugarcane bagasse (SCB) as a raw material for the production of colloidal LNPs via two simple solvent shifting methods, and they compared the effects of bagasse pretreatment with alkaline and organosolv on the size and performance of the LNPs. The results showed that spherical lignin nanoparticles were formed, with sizes that depended on the preparation technique. The average size of the alkaline-LNPs was between 115 and 300 nm, while that of the organosolv-LNPs was between 270 and 680 nm. All LNPs in aqueous suspension had zeta potentials between -25 and -35 mV, a range that is stable for colloidal systems.

In this work, an agricultural residue, specifically rice husks, was used to produce LNP, MAS, and MS. A novel method isolated these nanoparticles, allowing for sustainable production without scarifying lignin. The Aluminium-assisted precipitation method demonstrates strong sustainability potential, particularly due to the absence of lignin loss, which preserves the intrinsic value of the biomass and avoids additional recovery steps. The process operates under mild conditions, indicating relatively low energy consumption compared to conventional separation techniques. In terms of scalability, the method is promising due to its simplicity and compatibility with existing processing systems.